Subsea Pipeline Free Span and VIV Fatigue Assessment

A Standards-Based Methodology for Identifying Free Spans and Managing Vortex-Induced Vibration Fatigue

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DNV-RP-F105 · DNV-RP-F114 · DNV-RP-C203

> This paper describes Cronus's general engineering methodology. Figures, dimensions, and results in the illustrative example are hypothetical and do not represent any actual client, field, or pipeline.

1. Introduction and Purpose

A subsea pipeline rarely rests on a perfectly even seabed. Wherever the seabed profile dips faster than the pipeline can settle into it, at a channel crossing, a fault scarp, an existing line crossing, or simple local unevenness, a length of pipe is left unsupported, forming a free span. Left alone, a free span exposed to a steady bottom current can lock into vortex shedding behind the pipe and begin to oscillate. Sustained over months and years, that oscillation is a high-cycle fatigue mechanism that can crack a girth weld long before the pipeline's design life is reached.

Cronus has developed and repeatedly applied a structured, standards-based methodology for identifying free spans along a subsea pipeline route, establishing how long a span can safely be left unmitigated, and specifying vortex-induced vibration (VIV) suppression only where it is actually required. This white paper describes that methodology: the engineering basis it requires, the analytical framework it follows, the fatigue criteria it evaluates, and the deliverables a client receives.

The methodology is illustrated in Section 7 using a hypothetical example pipeline. That example, including all dimensions, span lengths, and results, is fictitious and constructed solely to demonstrate how the framework is applied; it is not drawn from, and does not represent, any actual Cronus project or client.

2. Governing Codes and Standards

Cronus performs the free span and VIV fatigue assessment in accordance with the following industry codes and recommended practices, supplemented by any project- or client-specific line pipe and engineering specifications. Where a client specification and an industry code differ, the higher-order, more conservative document governs.

Standard	Title	Role in the Assessment
API Spec 5L	Specification for Line Pipe	Establishes line pipe manufacturing and mechanical property requirements used as the basis of the pipe's stress-strain inputs.
DNV-ST-F101	Submarine Pipeline Systems	Governing submarine pipeline design code, referenced for wall thickness, material, and safety-class requirements.
DNV-RP-F105	Free Spanning Pipelines	Primary basis for the free span screening approach, the VIV response model, and the allowable span length framework used throughout this methodology.
DNV-RP-F114	Pipe-Soil Interaction for Submarine Pipelines	Basis for the soil resistance and dynamic soil stiffness values used at the span shoulders.
DNV-RP-C203	Fatigue Design of Offshore Steel Structures	Basis for the S-N fatigue approach used to convert stress cycles into fatigue damage.

3. Engineering Basis: Required Input Data

A free span assessment is only as reliable as the seabed and environmental data behind it. Cronus compiles the categories of data below at the outset of every assessment, drawing on client-furnished engineering data, route and geohazard surveys, and Cronus's own pipe-soil interaction calculations.

Input Category	Description	Typical Source
Route and bathymetry	A KP-referenced digital terrain model of the seabed along the pipeline centerline, at a resolution fine enough to resolve individual spans.	Route and geophysical survey
Pipe properties	Outer diameter; wall thickness by segment; material grade; SMYS, SMTS, elastic modulus; corrosion allowance (fatigue calculations conservatively use half the design corrosion allowance).	Line pipe specification, mill data
Ancillary items	Buoyancy modules and other installation aids that change the pipeline's effective weight, added mass, and span shoulder stiffness.	Installation engineering
Geotechnical data	Soil classification along the route, generalized into representative soil provinces that set the dynamic soil spring stiffness at each span shoulder.	Geotechnical survey (per DNV-RP-F114)
Environmental / metocean data	Extreme (1-, 10-, 100-year) and long-term bottom current data by water depth, plus extreme wave data in water shallow enough for waves to influence VIV.	Metocean report
Installation (lay) tension	The planned bottom-tension profile, which sets the effective axial force retained in the pipeline and therefore its span stiffness.	Installation contractor / pipelay analysis
Pipe-soil interaction (PSI)	Vertical, axial, and lateral soil resistance at the span shoulders, which govern how deeply the pipe embeds and how a span's effective length develops.	Geotechnical / PSI analysis

The global route simulation described in Section 4 is deliberately run with the input combination that maximizes predicted span length: upper-bound lay tension paired with the soil and pressure/temperature conditions that minimize pipe embedment. The result is a conservative, not a best-estimate, picture of the as-laid route.

4. The Cronus Free Span & VIV Assessment Methodology

Cronus identifies and qualifies free spans in a five-stage process, aligned with DNV-RP-F105. The process separates two questions that are easy to conflate: where are the spans, and how long can a span at a given location safely be? The first question is answered with a full-route global model; the second is answered once, efficiently, for a handful of representative pipe/soil/environment combinations, and then applied everywhere that combination occurs.

Stage A: Global Route Simulation

A full-route finite element model of the pipeline, seated on the true 3-D seabed terrain, is used to predict the location, length, and gap of every free span along the route under the conservative input combination described in Section 3. This model also identifies which of those spans lie close enough to one another to interact dynamically.

Stage B: Allowable Span Length Derivation

The pipeline route is grouped into a manageable number of wall-thickness / geotechnical-province / metocean combinations (“groups”). For each group, isolated-span finite element models are built on an idealized flat seabed across a range of span lengths; modal analysis extracts the mode shapes and natural frequencies for each length, which are then run through a dedicated VIV fatigue tool to calculate factored fatigue damage against the DNV-RP-F105 response model. The allowable span length for that group is the longest length at which both long-term (design-life) and short-term (100-year extreme event) factored fatigue damage remain within budget.

Stage C: Span Screening

Every span identified in Stage A is compared against the allowable span length for its group. A span shorter than its group's allowable length requires no further action. A span that exceeds it is flagged for VIV suppression.

Stage D: Interacting Span Confirmation

DNV-RP-F105 recognizes that spans close enough together to interact dynamically generally exhibit more damping, and so less fatigue damage, than an isolated span of the same length. Cronus does not assume this credit; instead it screens the Stage A spans for interaction and, for a representative subset, builds a refined finite element model (matching the true local seabed shape, at true 2.5-D or 3-D fidelity, including any nearby crossing structures) and re-runs the modal and fatigue analysis. Interacting-span groups that confirm acceptable fatigue damage this way are not straked, even where an isolated-span check alone would have flagged them.

Stage E: Mitigation Design

For every span that still exceeds its allowable length after Stage D, Cronus specifies a VIV suppression strake: a triple-start helical strake, sized in pitch and height as a fraction of pipe diameter, with coverage sufficient to suppress vortex shedding along the flagged length. Strake locations at major crossings and geohazards are cross-checked against any adjacent, previously-qualified pipeline for consistency.

5. Fatigue Criteria

Fatigue damage from vortex-induced vibration is calculated using an S-N approach consistent with DNV-RP-C203, converting the stress range at each node into a number of cycles to failure, then summing the resulting damage across all significant modes and both in-line and cross-flow response using a linear (Palmgren-Miner) damage rule.

Assessment	Exposure Basis	Purpose
Long-term VIV fatigue	Design life (plus a margin for temporary/pre-operational exposure), driven by the long-term current probability distribution at each span's water depth.	Captures cumulative fatigue damage from ordinary operating currents over the full pipeline life.
Short-term VIV fatigue	A defined-duration 100-year extreme current (and, in shallow water, wave) event.	Confirms that even a single severe storm or current event does not consume an outsized share of the fatigue budget.
Interacting-span fatigue	Same long-term/short-term basis, applied to a refined multi-span model at true local bathymetry.	Confirms whether the damping credit from span interaction is sufficient to avoid strakes an isolated-span check would otherwise require.

The factored fatigue damage is calculated as D = n / Life, where n is the exposure (in cycles or years, as applicable) and Life is the factored fatigue life returned by the VIV fatigue tool. A span is deemed acceptable where the resulting factored damage remains below a defined allowable budget, set conservatively low and consistent with a medium safety class, to ensure no appreciable fatigue accumulates at spans left unmitigated.

6. Finite Element and Simulation Approach

Cronus uses a purpose-built combination of tools at each stage of the assessment, which provides an independent cross-check between the full-route prediction and the detailed fatigue calculation:

- A global pipeline FE tool driven by the true 3-D seabed surface (e.g., Fugro SAGE Profile) for the Stage A full-route span prediction and span-interaction screening.
- A general-purpose FE solver (e.g., Abaqus) for the Stage B isolated flat-seabed modal analysis across pipe/soil/environment groups, and for the Stage D refined interacting-span models built on the true local seabed shape.
- A dedicated multi-mode VIV fatigue tool (e.g., DNV's FatFree), which takes the mode shapes and frequencies from the FE modal analysis and, following the DNV-RP-F105 response model, returns factored fatigue damage for both in-line and cross-flow response, aggregated at each location.

Across all models, both in-plane and out-of-plane current directions are evaluated, all participating vibration modes are retained as contributing modes (rather than only the first mode), and dynamic soil spring stiffness at the span shoulders is calculated per DNV-RP-F114 rather than assumed rigid. Each of these choices materially affects the predicted natural frequency, and therefore the fatigue result.

7. Illustrative Worked Example

The pipeline described below, referred to as "Pipeline Y," along with all dimensions, span lengths, and results, is entirely hypothetical. It does not represent any actual Cronus client, project, or field, and no figures in this section should be read

as data from a real assessment. It is included solely to show how Sections 3–6 are applied in practice and what a resulting deliverable looks like.

7.1 Example Pipeline and Groups

Pipeline Y is a hypothetical 18-inch OD gas export pipeline, approximately 95 km in length, crossing several seafloor channels and fault scarps between a subsea drill center and a fixed platform. For illustration, the route is divided into four wall-thickness / geotechnical / metocean groups, summarized below.

Group	Wall Thickness [in]	Applicable KP Range [km]	Water Depth Context	Current Regime
#1	1.125	0 to 58.0	Deep water (no wave influence)	Metocean B
#2	1.062	58.0 to 89.5	Mid water (no wave influence)	Metocean C
#3 – 220 m WD	1.062	89.5 to 91.0	Shallow (wave-influenced)	Metocean D
#3 – 120 m WD	1.062	91.0 to 92.5	Shallow (wave-influenced)	Metocean D
#3 – 65 m WD	1.062	92.5 to End	Shallow (wave-influenced)	Metocean D

7.2 Allowable Span Length Results (Operation Condition)

Group	Short-Term Allowable [m]	Long-Term Allowable [m]	Governing Allowable [m]
#1	>140	135	135
#2	>92	90	90
#3 – 220 m WD	58	61	58
#3 – 120 m WD	42	47	42
#3 – 65 m WD	36	40	36

Consistent with the underlying physics, long-term fatigue governs the allowable span length in deep water while short-term (extreme-event) fatigue governs in shallow water, where wave action adds to the current-driven response.

7.3 Span Screening and Interaction Results

The Stage A global simulation predicts 96 free spans along the Pipeline Y route. Comparing each against its group's governing allowable length, 9 spans exceed the allowable and are flagged for mitigation. Of those 9, two span groups also screen as potentially interacting under DNV-RP-F105 and are re-evaluated with refined 2.5-D models; both confirm factored fatigue damage below 2% and are therefore not straked on that basis. The remaining 7 locations are recommended for VIV suppression strakes, for a total illustrative strake length of approximately 3,050 m.

Category	Count	Total Length	Disposition	Note
Spans identified (Stage A)	96	—	—	Full-route conservative prediction
Spans within allowable length	87	—	Complete, no action	No appreciable fatigue
Spans exceeding allowable, isolated basis	9	—	Flagged	Proceed to interaction screening
Interacting groups confirmed acceptable	2 groups	—	Complete, no strakes	Damage < 2% at both
Locations recommended for strakes	7	~3,050 m	Mitigate	Final recommendation

Counts and lengths above are illustrative only.

8. Typical Deliverables and Client Value

Deliverable	Content
Allowable span length matrix	Governing short-term and long-term allowable span length for every wall-thickness / geotechnical / metocean group along the route.
Span screening summary	Every predicted span, its length and location, and its disposition (acceptable / requires mitigation).
Interacting span report	Refined-model fatigue results for span groups qualifying for a span-interaction credit.
Strake specification	Recommended strake configuration (pitch, height, coverage) and length at every location requiring mitigation.
Conclusions and recommendations	Pass/fail summary by group and location, with supporting rationale and any survey/monitoring recommendations.

Where This Methodology Creates Value

- Allowable span lengths are derived once per group and applied route-wide, rather than re-analyzed span by span: a significant efficiency gain on routes with dozens or hundreds of predicted spans.
- The span-interaction credit in Stage D is claimed only where a refined model actually confirms it, avoiding both unnecessary strake cost and unsubstantiated risk reduction.
- Separating long-term and short-term fatigue checks ensures the governing mechanism, whether cyclic current fatigue in deep water or an extreme storm event in shallow water, is the one that actually sizes the mitigation.
- A defined, conservative fatigue budget for unmitigated spans keeps the pass/fail line unambiguous and auditable for the client's integrity management records.

9. Conclusion

Cronus has applied this free span and VIV fatigue methodology across multiple subsea gathering, export, and flowline routes spanning a range of water depths, seabed conditions, and pipe sizes. The framework is deliberately structured to answer the span-location question and the span-length question with the right tool for each, and to claim mitigation credit only where a refined model actually substantiates it.

We welcome the opportunity to discuss how this methodology applies to your project. Talk to an engineer, call 713.787.9444 or write to info@cronustec.com.